

Two-Level Hypergraph-Guided Multi-Agent Reinforcement Learning for Joint RB Allocation and Power Control in Underlay P2P HetNets

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Abstract—Underlay peer-to-peer (P2P) communication improves spectrum reuse in heterogeneous networks (HetNets), but couples resource allocation across links through shared radio endpoints and co-channel interference. Pairwise representations do not explicitly preserve these link groups, while a single uniform hypergraph processes endpoint sharing and resource-block (RB) reuse without distinguishing their physical origins. The two relations also differ in the information available before scheduling. Endpoint associations are relatively stable within each base station (BS), whereas current-slot RB assignments are unknown until actions are selected. This paper proposes hierarchical hypergraph-enhanced duplex dueling multi-agent Q-learning (HHG-QPLEX) for joint uplink RB allocation and discrete power control. The first hypergraph level represents intra-BS transceiver sharing, and the second represents cross-BS RB reuse using previous-slot assignments. A two-stage hypergraph neural network extracts the corresponding features and combines them with local observations for QPLEX under centralized training with decentralized execution (CTDE). Ablation and network-scaling experiments with 24 to 72 P2P agents show higher training return, sum rate, and peer-to-infrastructure (P2I) coverage than the evaluated QPLEX baselines with no structural features, pairwise graphs, or uniform hypergraphs.

Index Terms—HetNets, MARL, HGNNs, resource allocation, P2P communication, spectrum sharing

I. INTRODUCTION

SIXTH-generation (6G) wireless systems must support dense deployments and diverse services with stringent quality-of-service (QoS) requirements under limited spectrum and energy resources [1]–[4]. Heterogeneous networks (HetNets) combine macrocell coverage with small cells to increase spatial reuse and offload traffic in densely populated areas [5]–[7]. Within these networks, underlay peer-to-peer (P2P) links reuse licensed uplink resource blocks (RBs) to improve spectral efficiency [8]–[10]. Representative settings include ultra-dense femtocell overlays with hybrid overlay/underlay sharing [5], cross-cell machine-type traffic in industrial Internet of Things (IoT) [11], and multi-link P2P systems where each station jointly selects reusable resource units and transmit power [9]. In these settings, each receiver is affected by the aggregate co-tier and cross-tier interference from multiple co-channel transmitters [12], [13]. Joint RB allocation and power

control must therefore account for how groups of links share spectrum and interfere with one another.

Increasing the transmit power of a P2P link can improve its desired signal while reducing the rates of cellular and P2P receivers using the same RB. Moving that link to another RB changes the affected receiver set, so the preferred power level also depends on the reuse decision. A useful allocation policy must balance the benefit to the scheduled link against the interference imposed on the rest of the network.

Optimization and game-theoretic methods can coordinate resource allocation effectively when accurate network models are available and the number of links is moderate. Coalition formation and matching games have been applied to coordinate femtocell clusters and assign subchannels in ultra-dense HetNets [5], [14], [15]. Graph partitioning and weighted matching formulations address interference-aware scheduling in vehicular and edge-assisted networks [16]–[18]. However, joint RB allocation and power control form a mixed-integer non-convex problem, and centralized optimization becomes more costly as the number of links increases [6], [11], [19]. Even computing the interference degree under physical signal-to-interference-plus-noise ratio (SINR) constraints is NP-hard in hypergraph-based formulations [13]. Mobility, fading, and changing RB assignments also make it difficult for fixed interference models to track network conditions [20]–[22]. Learning-based resource allocation addresses these difficulties by adapting decisions through interaction with the wireless environment [23]–[26].

Deep reinforcement learning (DRL) and multi-agent reinforcement learning (MARL) have been applied to spectrum allocation when each agent observes only part of the network state [25]–[29]. Under CTDE, agents use global information during training to learn coordinated policies and select actions from their individual observations during execution [9], [30], [31]. Conflict and interference hypergraphs have been coupled with Markov decision process (MDP) formulations for massive IoT and cyber-physical communications, where federated or actor-critic schemes accelerate convergence under shared objectives [24], [32]. Value-decomposition methods further facilitate multi-agent credit assignment by factorizing a joint action-value function into agent-level utilities under cooperative rewards [33]–[36]. Among them, duplex dueling multi-agent Q-learning (QPLEX) improves representational flexibility while retaining the individual-global-max (IGM) property for decentralized greedy execution [37]. However,

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raw physical measurements do not explicitly identify the groups of links that share radio endpoints or reuse the same RB, although these groups determine the interference faced by each agent [9], [38].

An interference measurement reports the combined effect observed at a receiver, but does not by itself identify the contributing link group. Value decomposition provides a way to learn coordinated actions from a team reward, yet the learner must still infer these relationships from its inputs. This motivates a structural representation that describes which links are coupled before their action values are estimated.

Graph-based learning incorporates network topology into resource allocation by exchanging features between links connected by proximity or pairwise interference relations [16], [39]–[41]. Supervised, unsupervised, and spatiotemporal GNN frameworks have been reported for link scheduling, beam selection, traffic-aware resource allocation, and power control in dense wireless systems [22], [40]–[43]. Pairwise edges do not explicitly identify the full set of co-channel transmitters contributing interference to a receiver [11], [12], [44]. In underlay P2P HetNets, links associated with the same base station (BS) may share a transmitter or receiver, while links in different cells may reuse the same RB. A hypergraph represents each such group with a hyperedge connecting all member links, making endpoint sharing and common RB reuse explicit in the network representation [10], [45]–[47]. Prior studies have applied hypergraph coloring, matching, and spectral clustering to dense non-orthogonal multiple access (NOMA) HetNets, ultra-dense access networks, and cross-cell industrial communications [6], [7], [11], [12], [32]. Directed and oriented hypergraphs further distinguish asymmetric or cumulative interference among multiple terminals [44], [48], [49]. Location-aware coloring has been used to restrict underlay reuse while protecting cellular QoS [8], [50].

Endpoint sharing and RB reuse describe different aspects of link coupling. Two links may share a transmitter or receiver even when they are assigned different RBs. Conversely, links with distinct endpoints may interfere because they reuse the same RB across cells. The first relation follows relatively stable connectivity, while the second changes with resource scheduling [4], [29]. Combining them in a common aggregation stage makes it harder to distinguish persistent endpoint relationships from the co-channel groups created by a particular allocation.

A hierarchical representation preserves this distinction. It first summarizes the endpoint-sharing neighborhood of each link, then uses the updated features to represent cross-cell RB reuse. The two stages also allow different aggregation rules. Endpoint-sharing groups can contain neighbors with unequal relevance to the reference link, making selective aggregation useful when hyperedges are large [51], [52]. The RB-sharing stage instead retains the complete historical reuse group to describe the set of links that occupied the resource.

RB-sharing features must also respect the information available before action selection. Current-slot RB assignments and transmit powers are determined by the policy, so they cannot be used to construct its input for the same decision. Dynamic hypergraphs for mobile satellite and socially driven

emergency networks likewise rely on observed network states to track interference over time [20], [21], [38]. Using previous-slot RB assignments provides an observable description of recent co-channel relationships. Combined with the more stable endpoint-sharing groups, this history supports a two-stage representation that first processes intra-cell connectivity and then cross-cell reuse.

The resulting features must support cooperative value learning. A shared reward evaluates the joint allocation but does not separate individual contributions, and concurrent policy updates change the interference conditions faced by each agent [31], [53]–[55]. QPLEX addresses credit assignment through attention-based value mixing while preserving IGM-consistent greedy execution [37]. Its value estimates still depend on the information encoded in the agent features. HGNN-based reinforcement learning has been used to extract interference features in multi-cell and satellite networks [20], [38], [51], and to represent higher-order coordination in MARL [54], [56], [57]. These representative designs use a single hypergraph construction without explicitly separating shared endpoints from historical RB reuse. HHG-QPLEX makes this distinction in the observation features supplied to QPLEX and retains the original value-decomposition mechanism.

The remaining gap is therefore an observation representation that jointly preserves the two link relations and respects the scheduling order. Higher-order connectivity identifies the members of a group, but the controller also needs to distinguish why those links are grouped and when that membership is known. For underlay P2P allocation, this requires endpoint-sharing features that remain available as RB assignments change, together with reuse features constructed from completed scheduling decisions. These requirements motivate the two-level representation developed in this paper.

This paper proposes hierarchical hypergraph-enhanced QPLEX (HHG-QPLEX) for joint uplink RB allocation and discrete power control in underlay P2P HetNets. The model uses intra-BS hyperedges to group links with shared endpoints and cross-BS hyperedges to group links assigned to the same RB in the previous slot. A two-stage HGNN first aggregates the endpoint-sharing features using scaled dot-product similarity and top- K selection, then processes the complete historical RB-sharing groups. Each agent combines the resulting embedding with its physical measurements as input to QPLEX under CTDE.

We evaluate training return, sum rate, and peer-to-infrastructure (P2I) coverage through ablation and network-scaling experiments with matched settings. The main contributions are as follows.

- We construct a two-level hypergraph that separates intra-BS transceiver sharing from cross-BS RB reuse. The RB-sharing hyperedges use previous-slot assignments, so both levels can be formed before the current allocation is selected.
- We develop a two-stage HGNN that processes endpoint-sharing groups before historical RB-sharing groups. Scaled dot-product scoring and top- K selection retain relevant local features in the first stage. The embeddings

augment agent observations for QPLEX under CTDE while preserving its IGM-consistent greedy execution.

- We evaluate the learned RB and power selections, isolate the effects of hierarchical separation, and examine the similarity measure and top- K settings. Comparisons with uniform-hypergraph, pairwise-graph, and structure-free QPLEX baselines show higher training return, sum rate, and P2I coverage across networks with 24 to 72 agents.

Section II presents the system model and resource allocation problem. Section III develops the hierarchical hypergraph and HGNN. Section IV describes the cooperative learning formulation and HHG-QPLEX training and execution. Section V evaluates the proposed method, and Section VI concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section defines the underlay uplink network and the interference relationships that couple P2P allocation decisions. We first describe the cellular and P2P links, then derive their received SINRs and achievable rates. These expressions lead to a joint RB allocation and power-control problem that maximizes the network sum rate while protecting cellular uplinks.

A. System Model

We consider the two-tier uplink HetNet in Fig. 1, where P2P links reuse cellular RBs and introduce both intra-cell and inter-cell interference. The BS set $\mathcal{B} = \{b_0, b_1, \dots, b_S\}$ contains one macro BS (MBS) b_0 and S small BSs (SBSs). The SBSs are deployed within the macrocell to increase spatial reuse and offload local traffic. The licensed spectrum contains F orthogonal RBs, indexed by $\mathcal{F} = \{1, \dots, F\}$, each with bandwidth W Hz. Cellular uplinks use orthogonal frequency division multiple access (OFDMA), so CUEs in the same cell are assigned different RBs.

Resource allocation is performed in slots $t = 1, 2, \dots, T$. Each P2P link selects one RB and one discrete transmit-power level per slot, while channel gains vary with mobility and fading. Slot indices are omitted from the following rate expressions and retained when needed to distinguish current measurements from previous-slot information.

Each BS $b_s \in \mathcal{B}$ serves cellular user equipment (CUE) links and underlay P2P links on the same uplink pool. Let $\mathcal{M}_{b_s}$ and $\mathcal{K}_{b_s}$ denote the CUE and P2P link sets under b_s , with $M_{b_s} = |\mathcal{M}_{b_s}|$ and $K_{b_s} = |\mathcal{K}_{b_s}|$. An external cellular scheduler fixes the CUE RB assignments and transmit powers. We optimize the P2P allocations under this given cellular load.

The serving BS identifies the cell associated with a P2P link, while its data transmission takes place directly between the two endpoints. The P2P controller selects resources within the given cellular schedule and does not change the CUE assignments. Consequently, cellular uplinks provide both a background interference source for P2P receivers and a set of protected receivers whose rates depend on underlay reuse. This distinction determines which variables are optimized and which quantities enter the allocation constraints.

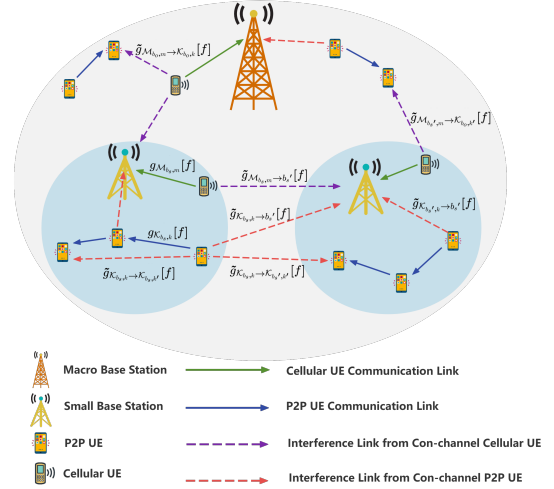


Fig. 1. Two-tier HetNet with underlay P2P reuse of cellular uplink RBs.

Each P2P link is identified by its transmitter-receiver pair and associated with one serving BS. Links associated with the same BS may share a transmitter or receiver, which couples their scheduling and interference conditions. Links in different cells also interact when they reuse the same RB. These two relations form the basis of the hierarchical model in Section III.

For example, links $\langle D_1, D_2 \rangle$ and $\langle D_1, D_3 \rangle$ belong to the same endpoint-sharing neighborhood because they have a common transmitter. A link $\langle D_4, D_5 \rangle$ in another cell has no such endpoint relation, but becomes a co-channel interferer when it reuses the same RB. Changing the RB assignment can remove the latter relationship without changing the former. The model therefore treats endpoint connectivity and RB-dependent interference as separate sources of information.

The channel power gain of the k -th P2P link under b_s on RB f is

$$g_{b_s,k}[f] = \alpha_{b_s,k} h_{b_s,k}[f], \quad (1)$$

where $h_{b_s,k}[f]$ is the small-scale fading coefficient and $\alpha_{b_s,k}$ captures large-scale path loss and shadowing. The binary indicator $\rho_{b_s,k}[f] \in \{0, 1\}$ marks whether RB f is assigned to P2P link k under b_s , and $P_{b_s,k}[f]$ is the corresponding transmit power. Likewise, $\rho_{b_s,m}[f]$ and $P_{b_s,m}[f]$ denote the fixed RB assignment and transmit power of CUE m under b_s on RB f .

Desired-link and interference-link gains play different roles in the allocation decision. The former determine the signal received by the intended receiver, while the latter determine the interference imposed on other receivers sharing the RB. Both are needed because a strong direct channel does not imply that a high-power transmission is favorable for the network. Power values in the SINR expressions are expressed in linear units, with the discrete dBm levels converted before evaluating received signal and interference powers.

Let $\tilde{g}_{a \rightarrow b}[f]$ denote the interference channel power gain from transmitter a to receiver b on RB f . Each BS or P2P receiver experiences interference from the other active

CUE and P2P transmitters using the same RB. The aggregate interference at BS b_s when decoding CUE m on RB f is

$$\begin{aligned}
 I_{b_s,m}[f] = & \sum_{k \in \mathcal{K}_{b_s}} \rho_{b_s,k}[f] P_{b_s,k}[f] \tilde{g}_{b_s,k \rightarrow b_s}[f] \\
 & + \sum_{\substack{b_s' \in \mathcal{B}, b_s' \neq b_s \\ m' \in \mathcal{M}_{b_s'}}} \rho_{b_s',m'}[f] P_{b_s',m'}[f] \tilde{g}_{b_s',m' \rightarrow b_s}[f] \\
 & + \sum_{\substack{b_s' \in \mathcal{B}, b_s' \neq b_s \\ k' \in \mathcal{K}_{b_s'}}} \rho_{b_s',k'}[f] P_{b_s',k'}[f] \tilde{g}_{b_s',k' \rightarrow b_s}[f].
 \end{aligned} \tag{2}$$

The aggregate interference at the receiver of P2P link k under b_s on RB f is

$$\begin{aligned}
 I_{b_s,k}[f] = & \sum_{m \in \mathcal{M}_{b_s}} \rho_{b_s,m}[f] P_{b_s,m}[f] \tilde{g}_{b_s,m \rightarrow b_s,k}[f] \\
 & + \sum_{\substack{k' \in \mathcal{K}_{b_s} \\ k' \neq k}} \rho_{b_s,k'}[f] P_{b_s,k'}[f] \tilde{g}_{b_s,k' \rightarrow b_s,k}[f] \\
 & + \sum_{\substack{b_s' \in \mathcal{B}, b_s' \neq b_s \\ m' \in \mathcal{M}_{b_s'}}} \rho_{b_s',m'}[f] P_{b_s',m'}[f] \tilde{g}_{b_s',m' \rightarrow b_s,k}[f] \\
 & + \sum_{\substack{b_s' \in \mathcal{B}, b_s' \neq b_s \\ k' \in \mathcal{K}_{b_s'}}} \rho_{b_s',k'}[f] P_{b_s',k'}[f] \tilde{g}_{b_s',k' \rightarrow b_s,k}[f].
 \end{aligned} \tag{3}$$

The BS interference expression contains contributions from local P2P transmitters, CUEs in other cells, and P2P transmitters in other cells. It contains no contribution from other CUEs in the same cell because their uplink RB assignments are orthogonal. A P2P receiver, in contrast, can experience interference from both the local CUE transmission and other P2P links, in addition to the corresponding inter-cell sources. In each sum, the RB indicator selects the active co-channel transmissions, and the interference gain determines their received contribution. Interference powers are added on the occupied RB without modeling coherent combining between independent transmitters.

With additive white Gaussian noise (AWGN) power σ^2 , the SINR of CUE m at BS b_s on RB f is

$$\gamma_{b_s,m}[f] = \frac{P_{b_s,m}[f] g_{b_s,m}[f]}{I_{b_s,m}[f] + \sigma^2}, \tag{4}$$

where $g_{b_s,m}[f]$ is the desired uplink channel power gain. The SINR of P2P link k under b_s on RB f is

$$\gamma_{b_s,k}[f] = \frac{P_{b_s,k}[f] g_{b_s,k}[f]}{I_{b_s,k}[f] + \sigma^2}. \tag{5}$$

Assuming Gaussian signaling and treating co-channel interference as noise, the achievable rate of CUE m served by BS b_s is

$$R_{b_s,m} = \sum_{f=1}^F \rho_{b_s,m}[f] W \log_2(1 + \gamma_{b_s,m}[f]), \tag{6}$$

and the achievable rate of P2P link k under b_s is

$$R_{b_s,k} = \sum_{f=1}^F \rho_{b_s,k}[f] W \log_2(1 + \gamma_{b_s,k}[f]). \tag{7}$$

Hereafter, the P2I rate refers to the cellular uplink rate $R_{b_s,m}$. The interference terms show that each P2P RB and power decision affects both cellular and P2P transmission rates. Interference resulting from the current joint allocation becomes observable only after that allocation is executed.

B. Problem Formulation

RB selection determines which links interfere, and power control determines the strength of their interference. Optimizing either variable in isolation leaves the other source of coupling unresolved. A low power level may protect a cellular receiver but unnecessarily limit a P2P link on a lightly interfered RB. A different RB may permit a higher power while causing less interference to protected uplinks. The allocation problem must therefore evaluate each RB and power combination through its effect on both P2I and P2P rates.

The design goal is to maximize the sum rate of all P2I and P2P links subject to P2P RB assignment, discrete power, and cellular QoS constraints. Let $\boldsymbol{\rho} = \{\rho_{b_s,k}[f]\}_{f \in \mathcal{F}}$ and $\mathbf{P} = \{P_{b_s,k}[f]\}_{f \in \mathcal{F}}$ collect the P2P RB and power variables. The P2I schedules $\{\rho_{b_s,m}[f], P_{b_s,m}[f]\}$ remain fixed background parameters. The optimization problem is

$$\begin{aligned}
 \mathbf{P1}: \max_{\boldsymbol{\rho}, \mathbf{P}} & \sum_{b_s \in \mathcal{B}} \left(\sum_{m \in \mathcal{M}_{b_s}} R_{b_s,m} + \sum_{k \in \mathcal{K}_{b_s}} R_{b_s,k} \right) \\
 \text{s.t.} \quad & \text{C1: } \rho_{b_s,k}[f] \in \{0, 1\}, \sum_{f=1}^F \rho_{b_s,k}[f] = 1, \\
 & \quad \forall b_s \in \mathcal{B}, k \in \mathcal{K}_{b_s}, f \in \mathcal{F}, \\
 & \text{C2: } P_{b_s,k}[f] \in \mathcal{P} = \{P_1, \dots, P_{L_p}\}, \\
 & \quad \forall b_s \in \mathcal{B}, k \in \mathcal{K}_{b_s}, f \in \mathcal{F}, \\
 & \text{C3: } R_{b_s,m} \geq R_{\min}^c, \\
 & \quad \forall b_s \in \mathcal{B}, m \in \mathcal{M}_{b_s}.
 \end{aligned}$$

Constraint C1 assigns one RB to each P2P link, and C2 restricts its transmit power to L_p discrete levels. Constraint C3 requires every cellular uplink to achieve at least $R_{\min}^c$. P2P reuse changes cellular interference even though the CUE allocations are fixed, so both link types appear in the sum-rate objective and cellular QoS is explicitly constrained.

Including cellular rates in the objective alone does not protect every cellular link. An increase in the rate of several P2P links can outweigh the rate loss of one cellular uplink in the sum, even when that uplink falls below its required service level. Constraint C3 therefore imposes a link-specific requirement in addition to the network objective. It separates aggregate efficiency from cellular service protection, which are evaluated through sum rate and P2I coverage in Section V.

Problem **P1** is mixed-integer and non-convex, with a joint decision space of size $\mathcal{O}\left((FL_p)^{\sum_{b_s} K_{b_s}}\right)$. The interference terms $I_{b_s,m}[f]$ and $I_{b_s,k}[f]$ couple the SINRs of co-channel

links, while mobility and fading change these couplings over time [58]–[60]. We address the problem through cooperative MARL under CTDE. Before introducing the learning formulation, Section III develops a representation of the link groups responsible for these couplings. Section IV then defines the decentralized partially observable MDP (Dec-POMDP), using a P2I-rate deficit penalty to account for C3. This penalty supports learning but does not guarantee that every cellular link satisfies C3 in every slot.

III. HIERARCHICAL HYPERGRAPH REPRESENTATION AND HGNN DESIGN

The interference model specifies how a joint allocation affects received rates, but does not provide the link-group features needed by the agents. This section develops that representation. After introducing hypergraph message passing, we construct separate hyperedges for intra-BS endpoint sharing and historical cross-BS RB reuse. We then define the two-stage HGNN that converts these relations and physical measurements into an embedding for each P2P link.

A. Hypergraphs and Hypergraph Neural Networks

A hypergraph extends an ordinary graph by allowing each hyperedge to connect an arbitrary finite set of vertices [61]–[64]. Formally, a hypergraph is denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is a finite vertex set and each hyperedge $e \in \mathcal{E}$ is a nonempty subset $e \subseteq \mathcal{V}$. When $|e| = 2$ for every e , the structure reduces to an ordinary undirected graph.

In underlay resource allocation, several co-channel transmitters contribute to the aggregate interference at a receiver [65]–[67]. Links can also share a radio endpoint within the same cell. Pairwise edges represent individual relationships but do not explicitly retain either group as a single relation [68]. A hyperedge identifies all members of the group and provides a common unit for feature aggregation [61], [62], [68]. This representation has been used for reuse coloring, clique-based allocation, and cross-cell coordination [6], [11].

An HGNN first aggregates vertex features within each hyperedge, then passes the resulting group features back to the incident vertices. Let $\mathbf{x}_v^{(l)}$ and $\mathbf{y}_e^{(l)}$ denote the features of vertex v and hyperedge e at layer l . These two operations are written as

$$\mathbf{y}_e^{(l)} = \phi_{v \rightarrow e}(\{\mathbf{x}_v^{(l)} \mid v \in e\}, e; \boldsymbol{\theta}_{v \rightarrow e}), \quad (8)$$

$$\mathbf{x}_v^{(l+1)} = \phi_{e \rightarrow v}(\mathbf{x}_v^{(l)}, \{\mathbf{y}_e^{(l)} \mid v \in e\}, v; \boldsymbol{\theta}_{e \rightarrow v}), \quad (9)$$

where $\phi_{v \rightarrow e}$ and $\phi_{e \rightarrow v}$ denote permutation-invariant aggregation operators, such as sum, mean, or max pooling, with learnable transformations parameterized by $\boldsymbol{\theta}_{v \rightarrow e}$ and $\boldsymbol{\theta}_{e \rightarrow v}$ [62]. Residual links and normalization are commonly inserted to stabilize training under heterogeneous hyperedge sizes [63], [69].

Hyperedge aggregation gives each vertex access to a feature summary of the group to which it belongs. Within a large endpoint-sharing group, however, equal weighting can obscure the features most relevant to the reference link. Locality constraints and top- K selection allow the update to retain the

most relevant neighbors. We next apply these operations to intra-BS endpoint sharing and use a separate aggregation stage for cross-BS RB reuse.

B. Hierarchical Hypergraph Construction 加图

We construct separate hyperedge sets for transceiver sharing and RB reuse. The first follows endpoint connectivity within each cell and changes with association or routing. The second follows RB assignments and is updated at each scheduling slot. Because current-slot assignments are not available before action selection, the RB-sharing set is formed from the previous allocation. The HGNN processes the two sets in sequence, using local endpoint-sharing features as input to cross-cell reuse aggregation.

The hierarchical hypergraph is $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{E}$ partitioned into intra-BS transceiver-sharing hyperedges and cross-BS RB-sharing hyperedges. Each vertex $v_{b_s, k} \in \mathcal{V}$ denotes the k -th P2P link under BS $b_s \in \mathcal{B}$ and is identified by the ordered transmitter-receiver pair $\langle D_x, D_y \rangle$.

Intra-BS hyperedges identify links whose channel and interference conditions are related through shared endpoints. This connectivity provides a structural prior for RB and power coordination. It does not impose an additional power constraint on the shared device. For each BS b_s and reference link c with $v_{b_s, c} = \langle D_x, D_y \rangle$, the transceiver-sharing hyperedge $e_{b_s, c}$ contains the reference link and all other P2P links in the same cell with at least one endpoint in $\{D_x, D_y\}$. Thus,

$$e_{b_s, c} = \{v_{b_s, c}\} \cup \{v_{b_s, k} = \langle D_i, D_j \rangle \mid k \neq c, D_i \in \{D_x, D_y\} \vee D_j \in \{D_x, D_y\}\}. \quad (10)$$

Each hyperedge is centered on a reference link c , allowing Stage 1 to score neighboring links according to their relevance to that link. Its membership depends on endpoint connectivity and remains unchanged when RB assignments vary.

Cross-BS RB-sharing hyperedges represent co-channel reuse interactions across cells. For each RB $f \in \mathcal{F}$, let $\rho_{b_s, k}^{t-1}[f] \in \{0, 1\}$ mark whether link k under BS b_s occupied RB f after the actions at time $t - 1$. The corresponding hyperedge is

$$e_{rbf} = \{v_{b_s, k} \mid b_s \in \mathcal{B}, k \in \mathcal{K}_{b_s}, \rho_{b_s, k}^{t-1}[f] = 1\}. \quad (11)$$

The hyperedge e_{rbf} records the most recent set of links that reused RB f . It is available before the current decision and is refreshed after the next allocation is executed. This history provides recent co-channel information, although the current actions may change the reuse group. In contrast, the transceiver-sharing set changes only when endpoint connectivity is updated.

Stage 1 aggregates features within the transceiver-sharing hyperedges $\{e_{b_s, c}\}$. Stage 2 then combines these updated features within the historical RB-sharing hyperedges $\{e_{rbf}\}$. Separate aggregation operators allow the two relations to receive different weights, while the sequential update carries local endpoint information into the representation of cross-cell reuse.

C. Hierarchical HGNN

The hierarchical HGNN (HHGNN) uses scaled dot-product scoring and top- K selection in Stage 1, followed by softmax-weighted aggregation of each complete RB-sharing group in Stage 2. In the second stage, the group mean is the reference for computing similarity scores. The final embedding $\mathbf{h}_{b_s,k}^{(L)}$ is combined with physical measurements to form the agent observation in Section IV.

Each vertex $v_{b_s,k}$ carries an initial feature $\mathbf{x}_{b_s,k}^{(0)}$ constructed from decision-time channel, interference, and BS-tier features. Let $\mathbf{q}_{b_s}^{\text{tier}}$ denote the one-hot serving-BS-tier vector, $\tilde{\mathbf{g}}_{b_s,k}^t$ denote the normalized P2P-to-BS interference-gain vector over all BSs and RBs, $\mathbf{i}_{b_s,k}^{t-1}$ denote the P2P-receiver interference vector measured after the previous slot, and $\mathbf{g}_{b_s,k}^t$ denote the current direct-link gain vector over all RBs. We **define**

$$\mathbf{x}_{b_s,k}^{(0)} = [\mathbf{q}_{b_s}^{\text{tier}} \parallel \tilde{\mathbf{g}}_{b_s,k}^t \parallel \mathbf{i}_{b_s,k}^{t-1} \parallel \mathbf{g}_{b_s,k}^t]. \quad (12)$$

A shared linear layer projects the input into a d -dimensional feature vector for hyperedge aggregation,

$$\mathbf{h}_{b_s,k}^{(0)} = \mathbf{W}_{in} \mathbf{x}_{b_s,k}^{(0)} + \mathbf{b}_{in}, \quad (13)$$

where $\mathbf{W}_{in}$ and $\mathbf{b}_{in}$ are shared across vertices.

Stage 1 operates independently within each BS over $\{e_{b_s,c}\}$. The neighbors in a large endpoint-sharing group can differ in their relevance to link c , so each receives a score before aggregation. Selecting the highest-scoring neighbors limits the mixing of less relevant features. For reference vertex $v_{b_s,c}$, the score of $v_{b_s,k} \in e_{b_s,c}$ is computed by a scaled dot product,

$$w'(v_{b_s,k} \mid v_{b_s,c}) = \exp\left(\frac{\mathbf{h}_{b_s,c}^{(l)\top} \mathbf{h}_{b_s,k}^{(l)}}{\tau}\right), \quad (14)$$

where $\tau > 0$ is a learnable temperature initialized to $\sqrt{d}$. The top- K_1 neighbors with the highest scores form the sampling set $\tilde{N}_{tr}(v_{b_s,c})$, and the scores are renormalized inside this set,

$$w(v_{b_s,k} \mid v_{b_s,c}) = \frac{w'(v_{b_s,k} \mid v_{b_s,c})}{\sum_{v_{b_s,j} \in \tilde{N}_{tr}(v_{b_s,c})} w'(v_{b_s,j} \mid v_{b_s,c})}. \quad (15)$$

The transceiver-sharing hyperedge feature is the weighted sum of the sampled neighbors,

$$\mathbf{y}_{b_s,c}^{\text{intra}(l)} = \sum_{v_{b_s,k} \in \tilde{N}_{tr}(v_{b_s,c})} w(v_{b_s,k} \mid v_{b_s,c}) \mathbf{h}_{b_s,k}^{(l)}. \quad (16)$$

The hyperedge-to-vertex update returns cluster information to $v_{b_s,c}$. Relevance between $\mathbf{h}_{b_s,c}^{(l)}$ and each candidate hyperedge feature $\mathbf{y}_{b_s,c}^{\text{intra}(l)}$ is again scored by scaled dot-product similarity. The top- K_2 hyperedges are retained in $\tilde{E}_{tr}(v_{b_s,c})$ and normalized to weights $w(e_{b_s,k} \mid v_{b_s,c})$. The Stage 1 embedding update is

$$\mathbf{h}_{b_s,c}^{\text{upd}(l)} = \sigma\left(\mathbf{h}_{b_s,c}^{(l)} + \mu \sum_{e_{b_s,k} \in \tilde{E}_{tr}(v_{b_s,c})} w(e_{b_s,k} \mid v_{b_s,c}) \mathbf{y}_{b_s,k}^{\text{intra}(l)}\right), \quad (17)$$

where $\mu \in (0, 1)$ scales the transceiver-sharing contribution and $\sigma(\cdot)$ is an element-wise nonlinearity that separates Stage 1

from Stage 2. Top- K selection reduces the number of aggregated messages, but scores are still computed for all candidate members. The similarity cost therefore depends on hyperedge size, as analyzed in Section IV-D.

Stage 2 aggregates the updated features $\mathbf{h}_{b_s,k}^{\text{upd},(l)}$ over $\{e_{rbf}\}$. After an allocation satisfying C1, each link belongs to one RB-sharing hyperedge. All members have used the same RB and jointly contributed to its co-channel interference. Stage 2 therefore retains the complete group and assigns softmax weights to its members. For each nonempty e_{rbf} , the mean feature is

$$\bar{\mathbf{h}}_{rbf}^{(l)} = \frac{1}{|e_{rbf}|} \sum_{v_{b_s,k} \in e_{rbf}} \mathbf{h}_{b_s,k}^{\text{upd},(l)}, \quad (18)$$

where $|e_{rbf}|$ is the hyperedge size. Softmax weights $w(v_{b_s,k} \mid e_{rbf})$ are formed from the scaled similarities $\mathbf{h}_{b_s,k}^{\text{upd},(l)\top} \bar{\mathbf{h}}_{rbf}^{(l)} / \tau$ inside e_{rbf} , and the RB-sharing hyperedge representation is

$$\mathbf{y}_{rbf}^{(l)} = \sum_{v_{b_s,k} \in e_{rbf}} w(v_{b_s,k} \mid e_{rbf}) \mathbf{h}_{b_s,k}^{\text{upd},(l)}. \quad (19)$$

The Stage 2 vertex update is

$$\mathbf{h}_{b_s,k}^{(l+1)} = \sigma(\mathbf{h}_{b_s,k}^{\text{upd},(l)} + \nu \sum_{f=1}^F \rho_{b_s,k}^{t-1}[f] \mathbf{y}_{rbf}^{(l)}), \quad (20)$$

where $\nu \in (0, 1)$ scales the RB-sharing contribution, and the sum over the RB index $f \in \mathcal{F}$ selects the unique previous-slot RB-sharing hyperedge containing $v_{b_s,k}$.

Stages 1 and 2 are repeated for $l = 0, \dots, L-1$. The final embeddings $\{\mathbf{h}_{b_s,k}^{(L)}\}$ contain both local endpoint-sharing information and recent cross-cell RB-reuse information. Section IV incorporates these features into the agent observations used for local action-value estimation and centralized value mixing.

IV. HHG-QPLEX FOR JOINT RB ALLOCATION AND POWER CONTROL

This section connects the hierarchical representation to the RB and power decisions in **P1**. We first formulate the cooperative decision process by specifying the agents, observations, actions, and shared reward. We then introduce the QPLEX value decomposition and explain how the HHGNN embeddings enter local action-value estimation and centralized mixing, as shown in Fig. 2. The training and execution procedures establish the order in which measurements, embeddings, and actions are computed. Finally, the complexity analysis identifies the operations that grow with the number of links and hyperedge members.

A. Dec-POMDP Formulation for Joint Resource Allocation

We formulate the slot-level RB and power decisions in **P1** as a cooperative Dec-POMDP. Each P2P link acts on its own observation, while the joint allocation determines interference and rates through the model in Section II-A.

The agent set $\mathcal{N}$ contains one agent per P2P link, with $N = |\mathcal{N}| = \sum_{b_s \in \mathcal{B}} K_{b_s}$. Agent $i \in \mathcal{N}$ selects an action from the shared finite set $\mathcal{A}$ with $|\mathcal{A}| = FL_p$. Each action indexes exactly one RB and one discrete power level, matching

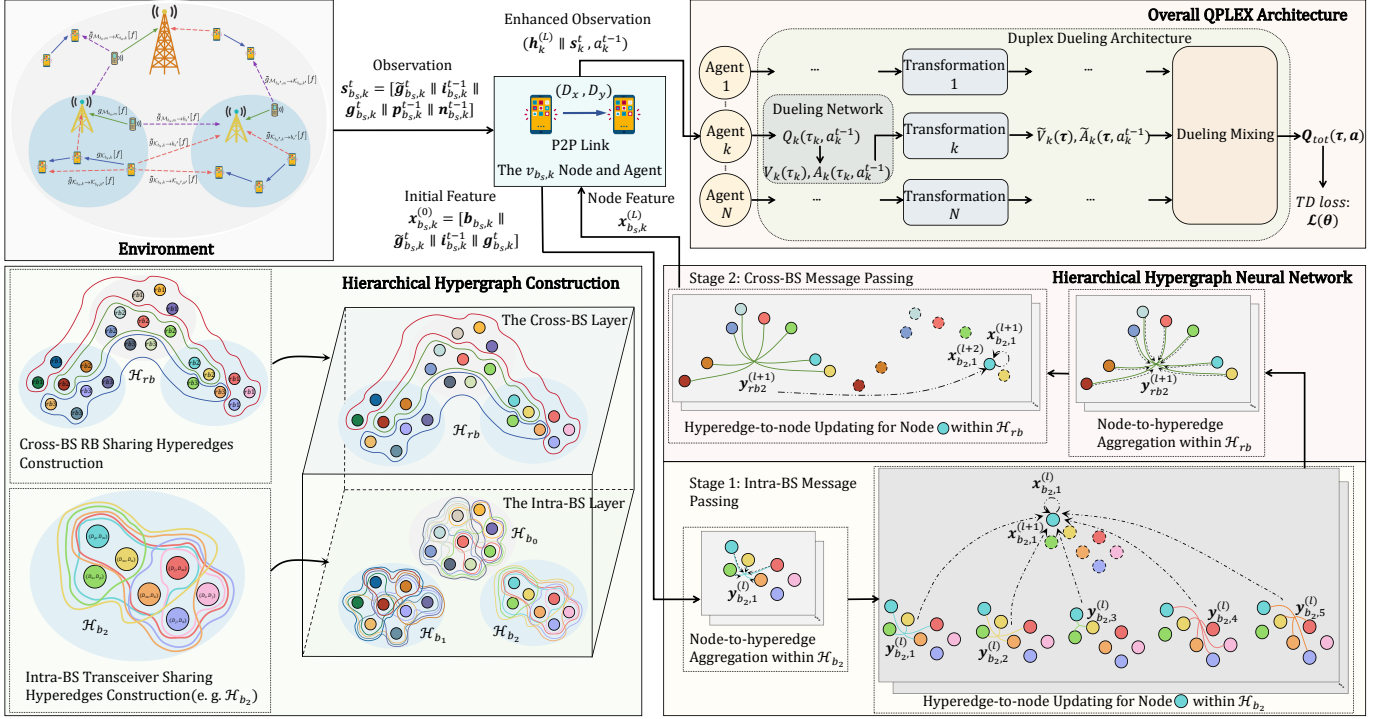


Fig. 2. HHG-QPLEX architecture connecting hierarchical hypergraph features with local action-value estimation and centralized QPLEX mixing.

constraint C1 and the discrete power set in C2. The joint action $\mathbf{a}_t = (a_{1,t}, \dots, a_{N,t})$ stacks all link decisions at interval t . State transitions follow the simulated mobility and fading processes. After the joint action is executed, the resulting co-channel interference enters the next observation. Thus, each agent's action changes the conditions observed by other links using the same RB.

The centralized trainer uses joint observations and actions to estimate the team action value. During execution, each agent selects its action independently from its observation and recurrent history. The observation includes the HHGNN embedding defined below.

The local observation of agent i at time t is denoted $o_{i,t}$. It concatenates raw physical measurements with the HHGNN embedding from Section III-C. Let b_s and k be the BS index and local link index of agent i under the simulator indexing map. The raw physical vector is

$$\mathbf{s}_{b_s,k}^t = [\tilde{\mathbf{g}}_{b_s,k}^t \parallel \mathbf{i}_{b_s,k}^{t-1} \parallel \mathbf{g}_{b_s,k}^t \parallel \mathbf{p}_{b_s,k}^{t-1} \parallel \mathbf{n}_{b_s,k}^{t-1}], \quad (21)$$

The components of $\mathbf{s}_{b_s,k}^t$ are the normalized P2P-to-BS interference gains $\tilde{\mathbf{g}}_{b_s,k}^t$ over all BSs and RBs, the prior-slot receiver-interference vector $\mathbf{i}_{b_s,k}^{t-1}$, the current direct-link gain vector $\mathbf{g}_{b_s,k}^t$, the prior-slot transmit-power pattern $\mathbf{p}_{b_s,k}^{t-1}$, and the binary same-cell RB-occupancy vector $\mathbf{n}_{b_s,k}^{t-1}$. Collectively, these components capture current channel conditions, interference levels, and recent allocation history.

The full local observation is

$$\mathbf{o}_{b_s,k}^t = [\mathbf{s}_{b_s,k}^t \parallel \mathbf{h}_{b_s,k}^{(L)}]. \quad (22)$$

The physical vector contains current channel gains and previous-slot interference and allocation measurements. The

embedding adds features aggregated from endpoint-sharing and historical RB-sharing groups. In particular, $\mathbf{n}_{b_s,k}^{t-1}$ records binary occupancy within the serving cell, **whereas Stage 2** represents the features of links that reused the same RB across cells.

When the learner consumes global information, the centralized state is $\mathbf{s}_t = \text{concat}(\mathbf{o}_{1,t}, \dots, \mathbf{o}_{N,t})$. Each agent also maintains an action-observation history $\tau_{i,t}$ for recurrent Q estimation.

All agents share one reward r_t aligned with the system objective in **P1**. Let $R_t^{\text{P2I}} = \sum_{b_s} \sum_m R_{b_s,m}$ and $R_t^{\text{P2P}} = \sum_{b_s} \sum_k R_{b_s,k}$ denote the total P2I and P2P rates at step t . Define the total P2I rate deficit

$$D_t^{\text{P2I}} = \sum_{b_s} \sum_m \max(0, R_{\min}^c - R_{b_s,m}), \quad (23)$$

where $R_{\min}^c$ is the cellular rate threshold in constraint C3. The shared reward is

$$r_t = \frac{1}{\eta} (\lambda R_t^{\text{P2I}} + \beta R_t^{\text{P2P}} - \alpha D_t^{\text{P2I}}), \quad (24)$$

where λ , β , and α are positive weights and η is a normalization constant. The first two terms reward P2I and P2P rates, while the deficit penalty discourages allocations that reduce a cellular rate below $R_{\min}^c$. The default setting uses $\alpha \gg \lambda, \beta$ to place greater emphasis on cellular QoS than on small rate gains. The factor η rescales the reward for temporal-difference (TD) learning. This formulation penalizes violations of C3 but does not enforce the constraint in every slot.

The cooperative objective maximizes the expected discounted return

$$J = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right], \quad (25)$$

where $\gamma \in [0, 1)$ is the discount factor. Episodes stop after a fixed horizon $T_{\max}$, so the simulated horizon is finite while the Bellman backup retains the standard form.

Because interference depends on all co-channel RB and power decisions, concurrent policy updates change the conditions faced by each agent. The shared reward evaluates the resulting allocation but does not identify individual contributions. We use QPLEX to learn the joint value from agent utilities and HHGNN features to represent the link relationships relevant to those utilities.

B. QPLEX Value Decomposition

Value-based MARL under CTDE factorizes a joint action-value function into per-agent terms so decentralized greedy choices stay consistent with the joint greedy action. The IGM property requires, for all joint histories τ ,

$$\arg \max_{\mathbf{a} \in \mathcal{A}^N} Q_{tot}(\tau, \mathbf{a}) = \left[\arg \max_{a_1 \in \mathcal{A}} Q_1(\tau_1, a_1), \dots, \arg \max_{a_N \in \mathcal{A}} Q_N(\tau_N, a_N) \right], \quad (26)$$

where Q_{tot} is the joint action value and Q_i is the local action value of agent i . Value decomposition networks (VDN) and QMIX satisfy IGM through additive and monotonic value factorization, respectively. QPLEX uses a duplex dueling construction to represent a broader class of joint values while retaining IGM [37]. This flexibility is useful when the effect of one link's action depends on the allocations of its co-channel neighbors. HHG-QPLEX retains this factorization and augments its observation inputs with structural features.

For each agent i , the local dueling split is

$$Q_i(\tau_i, a_i) = V_i(\tau_i) + A_i(\tau_i, a_i), \quad (27)$$

with

$$V_i(\tau_i) = \max_{a'_i \in \mathcal{A}} Q_i(\tau_i, a'_i). \quad (28)$$

During centralized training, a transformation network conditions the local dueling components on the joint history using positive weights $w_i(\tau) > 0$ and biases $b_i(\tau)$. The mixing module then forms Q_{tot} according to the QPLEX parameterization [37]. Training applies Bellman updates to this joint value, while IGM allows agent i to execute the local greedy action

$$a_i = \arg \max_{a \in \mathcal{A}} Q_i(\tau_i, a). \quad (29)$$

Training uses deep Q learning with replay memory $\mathcal{D}$. Each stored trajectory reconstructs joint histories τ , joint actions $\mathbf{a}$, rewards r , and next histories τ' . Let θ collect all trainable parameters, including recurrent encoders, the QPLEX transformation and mixing modules, and the HHGNN projection parameters updated end-to-end with the Q heads. Let θ^-

denote target parameters updated by soft Polyak averaging with coefficient $\tau_{\text{soft}} \in (0, 1)$. The TD loss is

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}} \left[\left(r + \gamma \max_{\mathbf{a}'} Q_{tot}(\tau', \mathbf{a}'; \theta^-) - Q_{tot}(\tau, \mathbf{a}; \theta) \right)^2 \right]. \quad (30)$$

The independent greedy actions form the joint allocation without a centralized search over $\mathcal{A}^N$.

C. HHG-QPLEX Training and Execution

At each scheduling slot, HHG-QPLEX forms the two hyperedge sets and computes an embedding for each P2P link. The embedding is concatenated with the physical observation and passed to the recurrent agent network for Q_i estimation. During training, the QPLEX transformation and mixing modules combine the agent values to estimate Q_{tot} , as shown in Fig. 2.

The structural features identify which neighboring links share endpoints and which reused the same RB in the previous slot. This information supplements the channel and interference measurements available to the value networks. The shared reward still provides the learning objective, while the two-stage representation makes the relevant link groups explicit in the inputs used for credit assignment.

The hypergraph at slot t uses endpoint connectivity and the stored RB indicators $\rho_{b_s, k}^{t-1}[f]$. The forward pass produces $\mathbf{h}_{b_s, k}^{(L)}$ and $o_{i, t}$ before evaluating the current action values. At episode reset, all previous-slot RB indicators are set to zero, so the initial observation contains no historical RB-sharing message.

Training proceeds in episodes. Each agent selects an RB and power level using ε -greedy exploration, and the simulator executes the joint allocation. The resulting P2I and P2P rates determine r_t , and the transition is stored for recurrent value learning. Once the replay buffer reaches the burn-in threshold, the learner samples episode minibatches, minimizes the TD loss, and softly updates the target parameters. Algorithm 1 gives the training procedure, and Algorithm 2 gives greedy evaluation.

Gradients from the TD loss update the QPLEX modules, recurrent encoders, and HHGNN projection and temperature parameters jointly. The structural representation is therefore trained with the action-value networks to capture features relevant to the allocation reward.

During evaluation, the parameters remain fixed and exploration is disabled. The evaluator constructs the hypergraphs and embeddings using the same information as in training, then each agent selects its greedy action. Performance is measured using the interference and rate expressions in Section II.

D. Complexity and Scalability Analysis

The trainable HHGNN components are the input projection $\mathbf{W}_{\text{in}}$, bias $\mathbf{b}_{\text{in}}$, and temperature τ . Hyperedge weights are computed directly from vertex features without separate query and key projections. Most trainable parameters therefore remain in the recurrent agent networks and QPLEX transformation and mixing modules.

In Stage 1, each reference vertex is scored against the members of its intra-BS hyperedge. For mean hyperedge

Algorithm 1 Centralized Training of HHG-QPLEX

- 1: Initialize HHGNN parameters, recurrent Q -network parameters, transformation and mixing parameters, replay buffer $\mathcal{D}$, and target parameters θ^-
 - 2: **for** each training episode **do**
 - 3: Reset the environment and recurrent hidden states, and set $\rho_{b_s,k}^0[f] \leftarrow 0$
 - 4: **for** time index $t = 1$ to $T_{\max}$ **do**
 - 5: Build intra-BS transceiver-sharing hyperedges and cross-BS RB-sharing hyperedges from topology and $\rho_{b_s,k}^{t-1}[f]$
 - 6: Compute embeddings $\mathbf{h}_{b_s,k}^{(L)}$ by the HHGNN forward map in Section III-C
 - 7: Form $o_{i,t} = [\mathbf{s}_{b_s,k}^t \|\mathbf{h}_{b_s,k}^{(L)}]$ and update histories $\tau_{i,t}$ for all agents
 - 8: Each agent selects $a_{i,t}$ by ε -greedy policy on $Q_i(\tau_{i,t}, a)$
 - 9: Execute $\mathbf{a}_t$, observe r_t and the next physical measurements, and store $\rho_{b_s,k}^t[f]$
 - 10: Append the transition $(\{\tau_{i,t}\}, \{a_{i,t}\}, r_t, \{\tau_{i,t+1}\})$ to $\mathcal{D}$
 - 11: **end for**
 - 12: **if** the replay buffer size reaches the burn-in threshold **then**
 - 13: Sample a mini-batch of episodes from $\mathcal{D}$
 - 14: Evaluate $\mathcal{L}(\theta)$ and apply one optimizer step to θ
 - 15: Soft-update targets by $\theta^- \leftarrow \tau_{\text{soft}}\theta + (1 - \tau_{\text{soft}})\theta^-$
 - 16: **end if**
 - 17: **end for**
-

Algorithm 2 Greedy Execution of HHG-QPLEX

- 1: Load trained parameters θ and disable exploration
 - 2: **for** each evaluation episode **do**
 - 3: Reset the environment and recurrent hidden states, and set $\rho_{b_s,k}^0[f] \leftarrow 0$
 - 4: **for** time index $t = 1$ to $T_{\max}$ **do**
 - 5: Build intra-BS and cross-BS hyperedges as in training
 - 6: Compute embeddings $\mathbf{h}_{b_s,k}^{(L)}$ by the same HHGNN forward map as in training
 - 7: Form $o_{i,t}$ and update $\tau_{i,t}$ for each agent i
 - 8: Each agent selects $a_{i,t} = \arg \max_{a \in \mathcal{A}} Q_i(\tau_{i,t}, a)$
 - 9: Execute the joint action, store the realized RB indicators $\rho_{b_s,k}^t[f]$, and obtain the next measurements
 - 10: **end for**
 - 11: **end for**
-

size $\bar{n}_e$ and feature dimension d , the similarity computation costs $\mathcal{O}(N\bar{n}_e d)$ per layer. Restricting membership to endpoint-sharing links within one BS avoids the $\mathcal{O}(N^2 d)$ cost of dense attention across all agents when $\bar{n}_e \ll N$. Top- K_1 and top- K_2 selection then restrict weighted aggregation to $\mathcal{O}(N(K_1 + K_2)d)$ operations. These budgets reduce the number of aggregated messages but do not remove the full similarity scan.

A conservative upper bound for Stage 2 is $\mathcal{O}(FNd)$ per

layer when each of the F RB groups is processed over all N links. Actual group aggregation depends on the occupied RBs and their membership sizes. Endpoint-sharing templates can be reused while connectivity remains fixed, whereas RB-sharing memberships are updated from $\rho_{b_s,k}^{t-1}[f]$. Replay storage grows with the number and length of stored episodes, and recurrent training memory grows with the episode batch size and unrolled horizon. Section V reports measured embedding times and allocation performance as the number of P2P links increases.

V. SIMULATION RESULTS AND ANALYSIS

The experiments examine whether the proposed representation improves allocation quality and which design choices account for the improvement. We first describe the simulation settings and comparison methods. Training returns and allocation statistics then show how the policy develops and which RB and power combinations it selects. Hierarchy ablations and similarity experiments separate the effects of the representation and aggregation rules. Finally, comparisons across network sizes assess the resulting sum rate and cellular coverage, together with the **measured embedding cost**.

A. Simulation Settings

We compare the methods using the same environment dynamics, reward in Eq. (24), discrete RB and power action space, and training budget. Each P2P link is one agent, and training runs for 4000 episodes of $T = 200$ steps. The Python and PyTorch implementation runs on a server with two Intel Xeon Gold 6530 CPUs, 512 GB DDR5 memory, and five NVIDIA L40 GPUs with 48 GB memory each.

The network contains one MBS with a coverage radius of 200 m and two SBSs with a radius of 50 m. Each P2P pair communicates within 25 m and maintains two links. **The numbers of RBs and cellular links increase with the P2P-link population across the evaluated network scales**. Large-scale fading uses the 3GPP TR 38.901 urban macro (UMa) and urban micro (UMi) path-loss models with log-normal shadowing [70]. Small-scale fading is Rician with $K = 10$ dB for line-of-sight (LOS) links and Rayleigh for non-line-of-sight (NLOS) links.

Each agent uses a single-layer gated recurrent unit (GRU) with hidden dimension 128. Parameters are updated by RMSProp with decay 0.99 and stabilizer $\epsilon = 10^{-5}$. The learning rate and exploration rate ε are linearly annealed from 5×10^{-4} to 5×10^{-5} and from 1.0 to 0.01 over 3200 episodes. Default Stage 1 budgets are $K_1 = 5$ and $K_2 = 5$. Table I lists the **remaining parameters**.

The rate plots report bandwidth-normalized aggregate rates in bit/s/Hz. With the common RB bandwidth W , this quantity is obtained by summing $R_{b_s,m}/W$ and $R_{b_s,k}/W$ over the cellular and P2P links. P2I coverage is the fraction of cellular uplinks whose rates meet the threshold $R_{\min}^c$,

$$C_t^{\text{P2I}} = \frac{\sum_{b_s \in \mathcal{B}} \sum_{m \in \mathcal{M}_{b_s}} \mathbb{I}\{R_{b_s,m}^t \geq R_{\min}^c\}}{\sum_{b_s \in \mathcal{B}} M_{b_s}}. \quad (31)$$

TABLE I: Network and Learning Parameters

Parameter	Value
Network	
Number of MBSs / SBSs	1 / 2
MBS / SBS coverage radius	200 / 50 m
Maximum P2P distance	25 m
Default P2I links per BS $M_{b,s}$	8
Default P2P links per BS $K_{b,s}$	16
Carrier frequency f_c	3.5 GHz
Per-RB bandwidth W	360 kHz
MBS / SBS antenna gain	15 / 10 dBi
BS noise figure	5 dB
UE antenna gain / noise figure	0 dBi / 7 dB
Noise power σ^2	-114 dBm
UE mobility speed	0.4 to 1.2 m/s
Path-loss model	3GPP TR 38.901 UMa/UMi
Power and reward	
P2P transmit power levels	{5, 10, 23} dBm
P2I transmit power	23 dBm
P2I rate weight λ	0.5
P2P rate weight β	0.5
P2I deficit penalty weight α	100
Reward scaling factor η	100
Learning	
Training episodes	4000
Steps per episode	200
GRU hidden dimension	128
Optimizer	RMSProp
RMSProp decay / stabilizer	0.99 / 10^{-5}
Initial / final learning rate	5×10^{-4} / 5×10^{-5}
Initial / final exploration rate	1.0 / 0.01
Discount factor γ	0.99
Soft update factor τ_{soft}	0.001
Replay buffer capacity	4200 episodes
Burn-in period	32 episodes
Minibatch size	32 episodes
HGNN	
Output embedding dimension d_{out}	32
Stage 1 neighbor top- K K_1	5
Stage 1 hyperedge top- K K_2	5

The two metrics describe different aspects of allocation quality. Aggregate rate measures the combined transmission benefit, while coverage measures how widely the cellular QoS requirement is satisfied. A rate increase accompanied by a coverage decrease therefore represents a different outcome from an improvement in both metrics.

HHG-QPLEX is compared with the following baselines under matched protocols.

- 1) QPLEX [37] uses the same value-decomposition framework without graph or hypergraph features. This comparison measures the benefit of structural observation augmentation.
- 2) GNN-QPLEX uses the pairwise connectivity construction in [39] to generate graph features for QPLEX. It provides a comparison between pairwise and hypergraph represen-

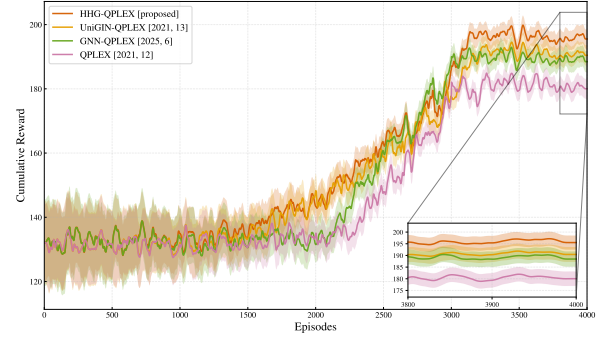


Fig. 3. Training returns of HHG-QPLEX and the QPLEX baselines under the default configuration. The inset compares late-training returns.

tations of link coupling.

- 3) Uni{GAT,GCN,GIN,SAGE}-QPLEX replaces the hierarchical module with a single-layer uniform hypergraph based on [71]. The variants use the same hyperedge construction with different aggregation operators, allowing us to assess a common hypergraph representation without separate processing of endpoint sharing and RB reuse. UniGIN-QPLEX is used as the strongest uniform-hypergraph reference in the subsequent comparisons.
- 4) Random selects an RB and a power level uniformly at each step. It provides a reference for the gain achieved by learned allocation policies.

B. Training Convergence

Fig. 3 compares how the structural representations affect learning progress under the reward in (24). All methods obtain low returns during early exploration, when frequent changes in RB and power selections produce poorly coordinated reuse and variable co-channel interference.

QPLEX improves more slowly and has the lowest late-training return. GNN-QPLEX performs better, indicating that pairwise link information is useful, but remains below UniGIN-QPLEX and HHG-QPLEX. The earlier return improvement of the two hypergraph methods supports the use of link-group features for learning coordinated allocations.

HHG-QPLEX also reaches the highest late-training return, as shown in the inset. Its advantage therefore extends beyond the initial learning phase. The hierarchy ablation below further examines whether separate processing of endpoint sharing and RB reuse accounts for this improvement.

C. Interference-Aware Allocation Behavior

Fig. 4 examines the allocation behavior behind the return improvement. HHG-QPLEX selects the lower power levels more frequently and uses 23 dBm only occasionally. The power-selection ratios stabilize as exploration decreases, while the RB-selection ratios remain unequal across the available resources.

The preference for lower power is consistent with the large P2I-deficit penalty, since reducing underlay transmit power lowers interference at the BSs for a given RB assignment.

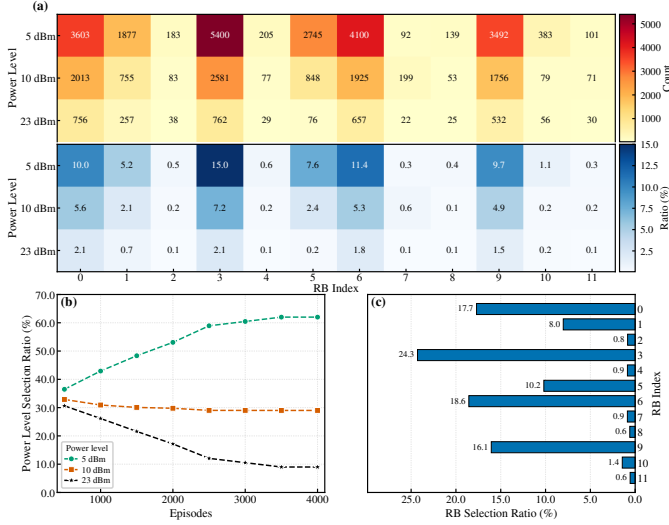


Fig. 4. Learned RB and power selections under the default configuration. (a) Counts and percentages of RB and power combinations. (b) Power-level selection ratios during training. (c) Overall RB selection ratios.

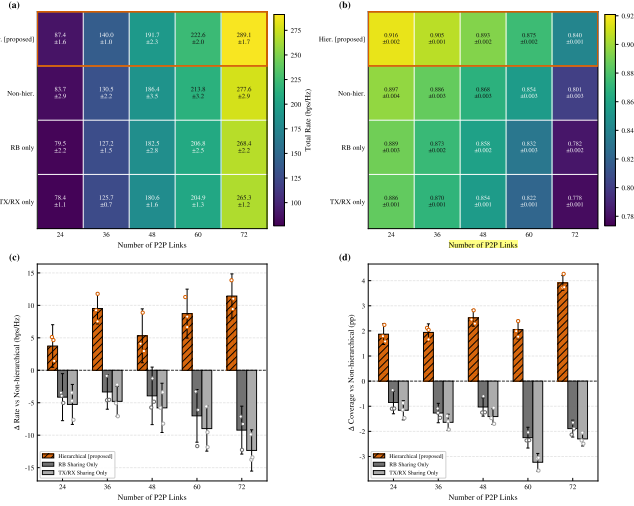


Fig. 5. Effect of hierarchical separation as the number of P2P links increases. (a) Sum rate and (b) P2I coverage of the hierarchical, non-hierarchical, RB-sharing-only, and transceiver-sharing-only models. (c) Sum-rate difference and (d) coverage difference relative to the non-hierarchical model. Error bars show the standard deviation across three seeds.

Unequal RB-selection ratios also show that the policy does not use all resources uniformly. Together, these statistics indicate selective RB reuse with conservative power control. They explain the policy's allocation tendency, while the following rate and coverage results assess its effect on network performance.

D. Effect of Hierarchical Separation

Fig. 5 separates the benefit of using both link relations from the benefit of processing them hierarchically. The full model outperforms the transceiver-sharing-only and RB-sharing-only variants, showing that the two relations provide complementary information. It also outperforms the non-hierarchical model, which includes both relations in a common aggregation

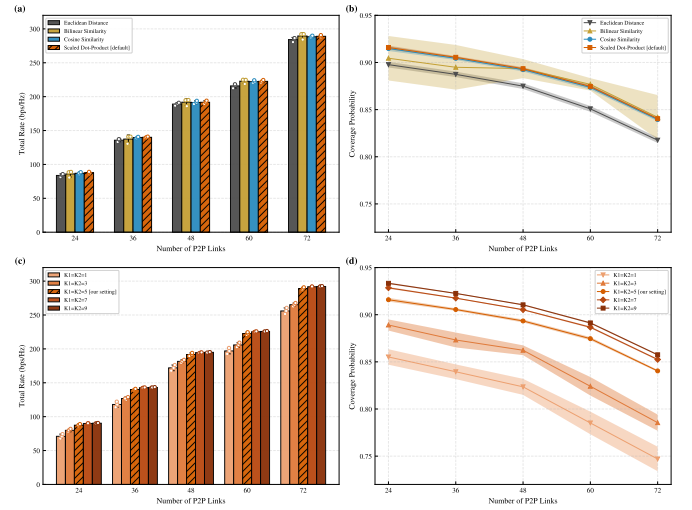


Fig. 6. Sensitivity to the similarity measure and Stage 1 selection budget. (a) Sum rate and (b) P2I coverage with four similarity measures applied in both HHGNN stages. (c) Sum rate and (d) P2I coverage with $K_1 = K_2 \in \{1, 3, 5, 7, 9\}$. Error bars show the standard deviation across three seeds.

stage. Thus, the gain comes from both the available structural information and how it is processed.

The hierarchical model achieves the highest sum rate and P2I coverage at every tested network size. Improving both metrics indicates that its aggregate-rate advantage is not accompanied by a loss of cellular coverage. The comparison with the non-hierarchical model is particularly informative because both models have access to endpoint-sharing and RB-sharing relations. Their performance difference therefore supports processing these relations separately.

This result is consistent with the order of feature aggregation. Endpoint-sharing aggregation first provides each link with information about its local connectivity. The subsequent RB-sharing stage combines these updated features within the historical co-channel group. A link can therefore receive reuse-related information that already reflects the endpoint relationships of the group members. The single-relation variants omit part of this context, while the non-hierarchical model combines the two relations without the same ordered processing. The ablation supports the value of this design, although it does not directly reveal which individual interference relationships determine each action.

E. Similarity Measure and Top-K Selection

Fig. 6 examines whether the allocation gains depend strongly on the similarity function and the number of selected neighbors. Cosine and scaled dot-product similarity achieve closely matched sum rates and P2I coverage, suggesting that both provide useful weights for aggregating the channel and interference features. Bilinear similarity achieves a competitive sum rate but shows greater variation in P2I coverage across seeds. Euclidean distance gives lower performance in both metrics. These results support cosine and scaled dot-product similarity for the evaluated feature representation, with embedding cost compared in Table II.

TABLE II: Average HHGNN Embedding Time for Different Similarity Measures and Network Sizes

Similarity measure	t_{hg} (ms)				
	24	36	48	60	72
Scaled dot-product	21.19	28.01	39.46	43.85	52.22
Cosine	18.46	27.19	37.72	45.26	55.84
Bilinear	20.96	29.21	35.36	44.63	57.81
Euclidean	19.83	26.56	35.14	43.99	54.31

Increasing $K_1 = K_2$ from 1 to 9 improves both metrics, but the gains become smaller beyond 5. We select $K_1 = K_2 = 5$ to retain most of the observed improvement while aggregating fewer messages than with 7 or 9 neighbors. This choice limits aggregation cost without claiming a reduction in the full similarity computation.

Table II examines the computational cost associated with the similarity choices. Embedding time increases with network size for all four measures, consistent with the need to process more link features and hyperedge memberships. Scaled dot-product similarity remains comparable in cost to the alternatives and has the lowest measured embedding time in the larger evaluated networks. Together with its rate and coverage performance, this makes it a suitable default for the proposed representation.

The timing results also clarify the role of selective aggregation. A fixed top- K budget limits the number of features entering the weighted aggregation, but candidate neighbors must still be scored before selection. Embedding time can therefore increase even when the selection budget remains unchanged. This agrees with the complexity analysis in Section IV and explains why a small aggregation budget does not make representation construction independent of network size. The measurements cover HHGNN embedding only, so they should be interpreted as the cost of structural feature extraction rather than the complete scheduling latency.

F. Performance across Network Scales

Fig. 7 examines whether the advantage of HHG-QPLEX persists across the evaluated network scales. Aggregate rate increases with network size, whereas P2I coverage declines. These trends describe different aspects of network operation. A larger aggregate rate does not by itself imply that cellular service requirements are more widely satisfied. The comparison between methods at each scale is therefore needed to assess how effectively the allocation policy balances transmission efficiency and cellular protection.

Under the matched settings at each scale, HHG-QPLEX achieves the highest sum rate and P2I coverage. Its advantage in both metrics suggests more effective coordination of RB reuse and transmit power under the shared reward. This observation is consistent with the allocation statistics, which show selective resource use and a preference for lower transmit powers. Those statistics alone do not establish the quality of an allocation, but the rate and coverage results show that the

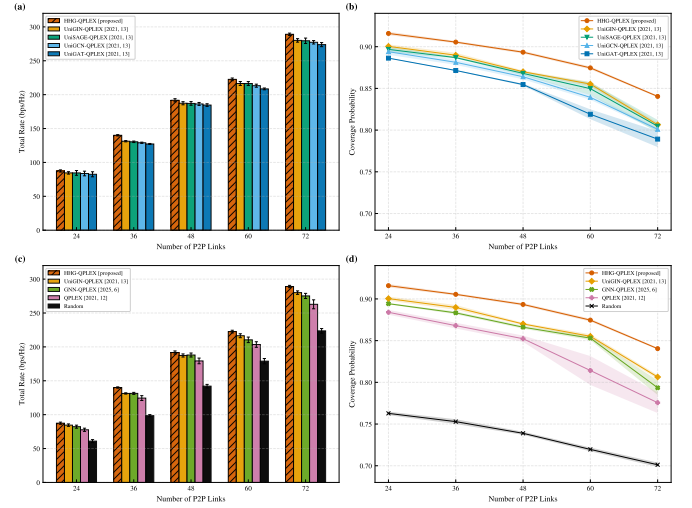


Fig. 7. Sum rate and P2I coverage as the number of P2P links increases. Panels (a) and (b) compare the uniform-hypergraph variants with HHG-QPLEX. Panels (c) and (d) compare HHG-QPLEX, UniGIN-QPLEX, GNN-QPLEX, QPLEX, and Random. Error bars show the standard deviation across three seeds.

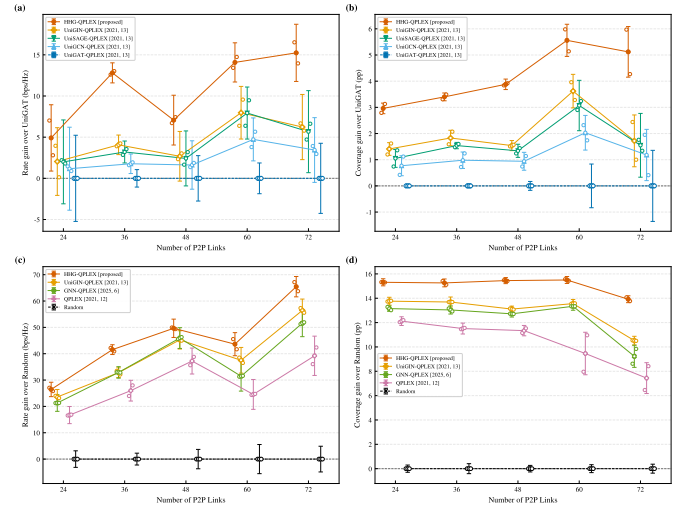


Fig. 8. Relative sum-rate and P2I-coverage gains as the number of P2P links increases. Panels (a) and (b) use UniGAT-QPLEX as the reference. Panels (c) and (d) use Random as the reference. Error bars show the standard deviation across three seeds.

learned allocation pattern is accompanied by better network performance.

The comparisons also distinguish the value of different structural representations. HHG-QPLEX outperforms the uniform-hypergraph variants, consistent with the hierarchy ablation in Fig. 5. Its additional advantage over GNN-QPLEX and QPLEX supports the use of explicit link-group features for RB and power coordination. The benefit remains visible at 72 links, where the coverage loss is most pronounced.

Fig. 8 expresses these differences relative to common reference methods. Panels (a) and (b) use UniGAT-QPLEX to compare the hypergraph models, while panels (c) and (d) use Random to measure the gain from learned allocation. HHG-QPLEX retains the largest relative gains across the tested sizes.

These gains vary with network size rather than increasing uniformly, so the result supports a consistent performance advantage over the tested range.

The comparisons with QPLEX and the structural baselines help explain where this advantage comes from. Without explicit link-group features, QPLEX must infer the relevant coupling from physical measurements and interaction histories. Graph and hypergraph augmentation provide additional context about the links associated with those measurements. The hierarchy further distinguishes endpoint connectivity from historical RB reuse, allowing the agent to use the two relations at different stages of feature extraction.

This distinction matters because the same interference level can arise from different combinations of active transmitters. The measured level describes the received interference, while structural features provide context about the links that may need coordinated decisions. Such context can help the value network distinguish allocation situations that appear similar in an aggregate measurement. The baseline and ablation results are consistent with this explanation. Together, they support using the hierarchical representation to inform joint RB and power decisions, while leaving the detailed interpretation of individual learned actions for further investigation.

VI. CONCLUSION

This paper proposed HHG-QPLEX for joint RB allocation and discrete power control in underlay P2P HetNets. The method separates intra-BS endpoint sharing from cross-BS RB reuse and constructs the latter from previous-slot assignments so that the representation is available before action selection. A two-stage HGNN selects relevant features within endpoint-sharing groups, then aggregates complete RB-sharing groups. The embeddings augment QPLEX observations while retaining its IGM-consistent greedy execution.

Experiments with 24 to 72 P2P links show earlier return improvement and higher late-training returns than the evaluated QPLEX learning baselines. HHG-QPLEX also achieves higher sum rates and P2I coverage than the evaluated learned and random allocation methods. The hierarchy ablation shows that both link relations are useful and that processing them separately improves performance over a common hypergraph aggregation stage.

Future work will extend the model to cross-cell P2P links with tier-dependent QoS requirements, examine hypergraph updates under mobility, and compare HHG-QPLEX with additional MARL methods.

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